



INTRODUCTION TO SUSTAINABILITY

LECTURE 13

ROLE OF TECHNOLOGY IN SUSTAINABLE DEVELOPMENT

- ♦ Scientists warn that our economies are already strained to the limit of the earth's capacity, and further growth, if unchecked, could very well lead to serious ecological imbalances and dangerous political instability.
- ♦ Today, many people remain optimistic, anticipating that the current economic boom will continue and that scientific and technological breakthroughs – information technology, biotechnology and others – will bring unparalleled economic growth to all parts of the world
- ♦ The road to the promising future that we desire can be found through science and technology. Science and technology can guide us to cost-effective approaches for improving environmental performance while, even more importantly, raising resource efficiency and profitability.



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SUSTAINABLE TRANSPORT



Transport drives development, linking people, connecting local communities to the world, building markets and facilitating trade. In turn, sustainable transport can drive sustainable development. The transport landscape varies greatly as one moves around the world and even within a given country or region. Almost everywhere in the developed world, people currently have transport options, though transport is not always available in an equitable or environmentally sensitive manner.

Accomplishing the SDGs will rely on advances in sustainable transport. Global progress in reducing greenhouse gas emissions cannot be realised without decisive action in sustainable transport, and countries cannot provide food security or healthcare without providing reliable and sustainable transport systems to underpin these advances.

ELECTRIC VEHICLE REVOLUTION!

Many industry observers believe we have already passed the tipping point where sales of electric vehicles (EVs) will very rapidly overwhelm petrol and diesel cars. Consumer behavior and awareness are changing as more people accept alternative and sustainable mobility modes. Industry players are accelerating the speed of automotive technology innovation as they develop new concepts of electric, connected, autonomous, and shared mobility. Electrification will play an important role in the transformation of the mobility industry and presents major opportunities in all vehicle segments. However, it is predicted that by **2035, the largest automotive markets will go electric!**





HYDROGEN POWERED VEHICLES

Hydrogen, the most abundant element in the universe, is increasingly viewed, along with electric vehicles, as one way to slow the environmentally destructive impact of the planet's 1.2 billion vehicles, most of which burn gasoline and diesel fuel.

Hydrogen fuel cells are cleaner and more efficient than traditional combustion-based engines and power plants. The benefits of Hydrogen fuel cells are: Reduced greenhouse gas emissions, High Reliability, Flexibility in installation and operation, Reduced demand for foreign oil and Improved environmental quality.

HYPERLOOP

First proposed in 2013 by Tesla and SpaceX as an open-source project free for anyone to develop, hyperloop is a theoretical transportation system that combines a near-vacuum effect with ultra-fast speeds. The magnetically-levitated, electrically-propelled pods will be able to transport passengers or freight through steel tubes at close to airline speeds of up to 1,223 kilometres per hour!

Hyperloop's sustainable technology can become part of the value chain. In other words, transportation through low-pressure tubes using levitated capsules propelled by electromagnetic propulsion has the potential to reduce energy consumption and CO2 emissions while increasing travel speeds. An additional value is that the hyperloop identified projects are not dependent on battery technology and can save this environmental footprint, among other things.





ADVANCEMENTS IN RENEWABLE ENERGY

Renewable energy resources make up 26% of the world's electricity today, but according to the IEA its share is expected to reach 30% by 2024. Renewable energy will continue to rise in the upcoming decade, edging out fossil fuels and reducing greenhouse gas emissions as their increasing deployment is crucial for efforts to tackle greenhouse gas emissions, reduce air pollution, and expand energy access

FLOATING SOLAR FARMS

Floating solar farms gained traction particularly in countries with high population density and with competing uses for limited available land. These also operate at high efficiency since installing the solar panels on water helps cool the equipment. Although electricity and water don't usually mix, a growing number of floating solar farms are being deployed worldwide.

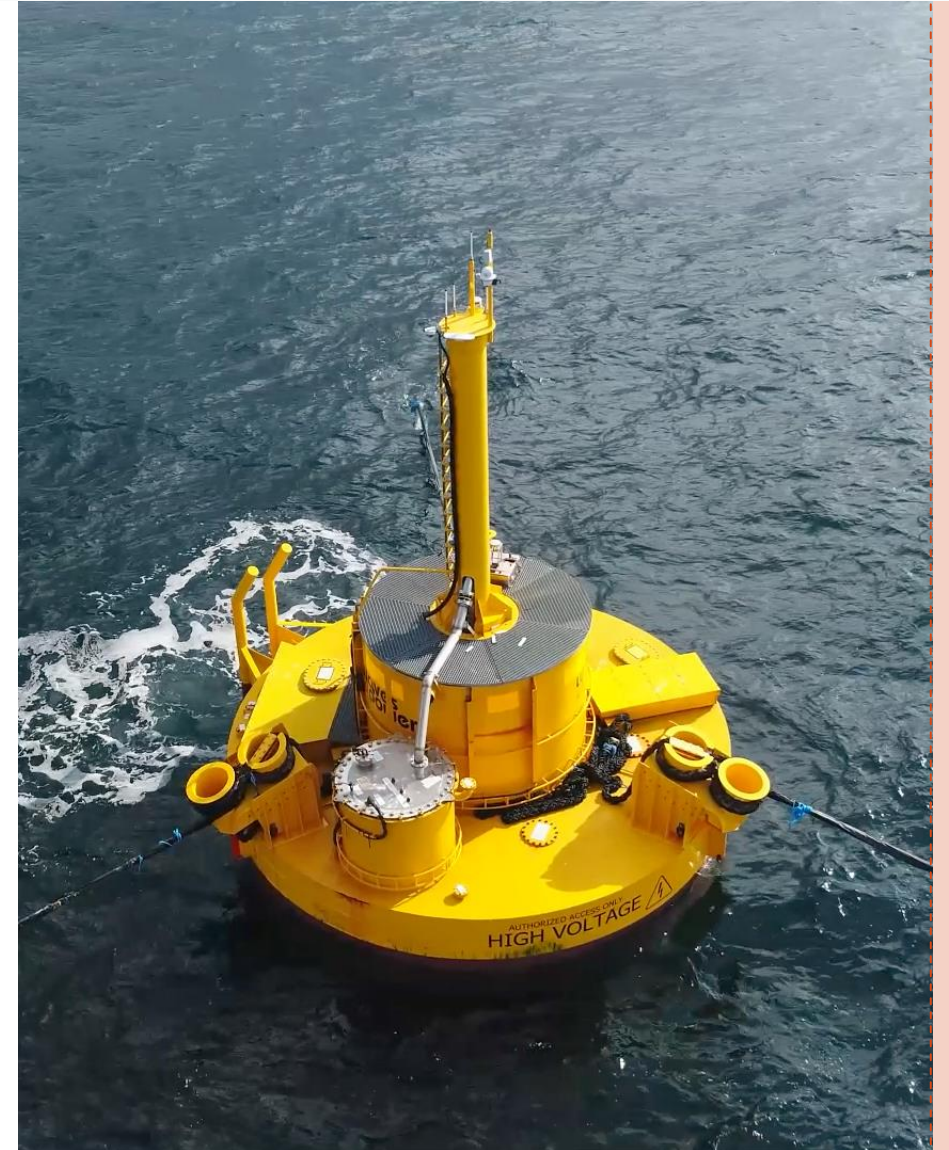
It is an eco-friendly method of generating electric power that combines marine technology and renewable energy technology. Solar modules are made to float on the surface of water, such as a dam or a reservoir. Electricity is sent through underwater cables to a transmission tower. As opposed to conventional rooftop solar panels, floating ones perform between 5% to 15% better because of the cooling effect of the water, and are not impacted by shading from other buildings



OCEAN ENERGY HARVESTING

Ocean wave energy is a very promising renewable energy source, and it is widely distributed worldwide. It is estimated that harnessing just 2 one-thousandths of the oceans' untapped energy could provide power equal to current worldwide demand.

Waves may be seasonal but are more constant—and more predictable—than wind or sunlight. Constancy and predictability enable a more straightforward and reliable integration into the electric utility grid. Wave energy also offers much higher energy densities, so devices can extract more power from a smaller volume at consequently lower costs.





SOLAR ROADWAYS

The ideology behind Solar Roadways was to implant solar panels into roads to produce electricity. The panels were allegedly also going to light up the roads with different LED patterns, replacing painted lines. For the winter, heating coils could melt snow and ice – all while generating electricity and requiring less maintenance.

This innovation however proved to be unsuccessful as none of the solar roadways installed as of today were cost-effective or efficient energy producers. The roads are expensive and produce far less electricity than what could be produced if the money was used on a solar farm – or by simply placing the panels by the side of the road.



FUTURE OF RECYCLING AND CIRCULAR ECONOMY

- The circular economy looks to prevent depletion of finite natural resources out of the global economy in order to better use the natural resources we've already extracted to extend their useful lives. Recycling is the gateway to greater sustainability in the circular economy. The future of the circular economy is bright. With technology innovating in this space at an ever-increasing rate, the way we think about waste and recycling is about to change.

TACKLING OCEAN POLLUTION

About eight million tonnes of plastic waste is added to the oceans every year — equivalent to one garbage truck of plastic being dumped into our oceans every minute! This figure is predicted to quadruple by 2050, if no action is taken. And plastic pollution is already having an increasingly devastating impact on our marine ecosystem.

Cleaning the ocean of plastics is a mammoth task. However, an innovative ocean-cleaning technology known as the **Seabin** is currently being trialed out to tackle this problem. Seabin collects trash floating in ports and marinas — and simultaneously collects data on the state of global waterways, guiding efforts to clean the oceans. The Seabin is also used to raise awareness and educate the public to prevent ocean pollution.



RECYCLED FASHION

- ♦ According to the [BBC](#), 92 million tons of textiles are wasted a year. Massive amounts of clothes end up in [landfills](#) and thus harming the planet.
- ♦ A number of fashion brands have stepped up with Recycled fashion initiatives to tackle this issue in the long run. Recycled Fashion refers to clothing made from previously used materials such as plastic bottles, nylon, polyester, and even old clothing.



ELIMINATION OF SINGLE USE PLASTICS

Single-use plastic packaging use is projected at 40.4 million tons in 2020 and is forecast to grow during 2020–2025 at a CAGR of 3.7% to 48.5 million tons in 2025. However, a number of new initiatives have been introduced to tackle this issue.

Edible Seaweed Pouches, Corn-Based Plastics, Bagasse (byproduct of sugarcane processing) and Mushroom Packaging are all alternative options that are being trialled out globally with the help of advanced technologies.



GLOBAL DISTRIBUTION OF SUSTAINABLE TECHNOLOGY



- ♦ The emergence of new technologies provides us with the means to deal with the challenges facing us today. However, unless the new technologies are adequately diffused and applied at the global level, their optimal effectiveness will not be attained. The challenge for us here today is how to enable the developing countries to properly utilise these technologies through **international co-operation**.
- ♦ In many cases, developing countries are simply **not technologically capable** of using energy efficiently and moving along the development path in harmony with nature. What is important is that we in the advanced countries have a realistic understanding of the needs of the developing countries and help them meet those needs.